A Deep Framework for Eddy Detection and Tracking From Satellite Sea Surface Height Data

Xin Sun, *Member, IEEE*, Meng Zhang, Junyu Dong, *Member, IEEE*, Redouane Lguensat, Yuting Yang, and Xirong Lu

***Abstract*— Ocean eddies, as a ubiquitous phenomenon of the global ocean, are extremely important for ocean energy and material exchanges. Therefore, efficient eddy detection and tracking are crucial for advancing our understanding of ocean dynamics. This work presents a framework for automatic ocean eddy detection and tracking by leveraging state-of-the-art machine learning algorithms. First, we propose a new convolutional neural network model for multieddies detection. This model is capable of extracting accurate boundary information of eddies and fitting the gap between semantic context and sea surface height (SSH). Second, a tracking algorithm is designed to track eddies lasting a number of days and provide visualization of the dynamical processes governing eddies' movements. Finally, we have made our data set publicly available, which is named SCSE-Eddy and can be used as a benchmark to evaluate the performances of artificial intelligence (AI)-based eddy detection methods. The data set covers daily remotely sensed SSH data located in the South China Sea and its eastern sea areas over a period of 15 years. The experimental results show that our methods achieve promising performances compared to existing approaches, especially for the eddies with indistinct geographical border. We believe that this work opens a new avenue for oceanographers to better discover and understand the physical properties of ocean eddies.**

***Index Terms*— Deep learning, eddy detection and tracking, ocean eddies, sea surface height (SSH).**

I. Introduction

OCEAN eddies are characterized by currents that flow in a roughly circular motion around the center of the eddy. In fact, ocean eddies exist in diverse rotating coherent structures, with diameters varying several hundred meters to hundreds of kilometers. Mesoscale eddies, whose radii are between 10 and 100 km, play a critical role in the distribution of nutrient and phytoplankton [1], [2], energy and salt transports [3], [4], and marine production [5]. Based on their rotation direction, mesoscale eddies can be split into two categories: cyclonic and anticyclonic eddies. Cyclonic eddies bring cold water, which is rich in nutrients and chlorophyll from the deep layers to the surface, thus affecting the distribution of photosynthetic phytoplankton, primary production, and the accumulation of fish [6]. In contrast, because of the opposite motion compared with cyclone eddies, anticyclonic eddies lead to an opposite effect by taking nutrients away and create a relatively unproductive nutrient-depleted surface layer. The existence and development of anticyclone eddies have a complete circulation system, which closely combines vertical motion with horizontal motion, and the center of the anticyclone is the downdraft [7].

Discovering and understanding the physics of ocean eddies are critical for both human activities and ocean sciences. However, eddy detection and tracking are still two challenging problems, especially for large geographic areas, presenting multiple eddies. This work devotes to developing a deep learning framework for automatic ocean eddy detection and tracking. The overall framework is shown in Fig. 1. Previous works that contributed to such topics could be divided into two categories: traditional algorithms [8]–[17] and artificial intelligence (AI)-based methods [18]–[22]. Traditional algorithms are primarily based on empirical physical parameters, thresh-

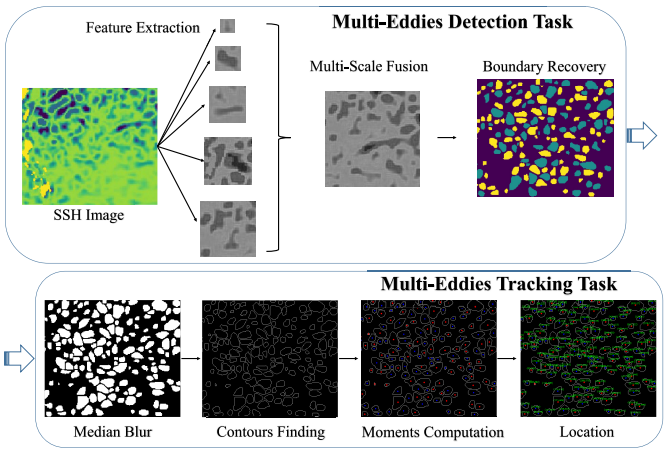


Fig. 1. Our deep learning framework consists of two successive tasks. The first task is to detect ocean eddies from the satellite remotely sensed SSH image. The second one is to track ocean eddies using the detection results.

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old values, and scientific computation (e.g., wavelet decomposition [16] and winding angle [17]). Meanwhile, AI-based methods employ powerful machine learning techniques to discover patterns from large data sets [23]. Recently, numerous deep learning techniques offer the advantage of inferring semantic information from remote sensing images [24]–[27]. The features extracted after the learning process are profitable to the interpretation of data [28]–[30] (e.g., text, image, and audio). Similar to many science and engineering tasks, the fields of object detection and semantic segmentation have also benefited from the recent success of deep learning models.

Traditional eddy detection and tracking algorithms [1], [8], [10] are sensitive to many factors, such as the distribution of chlorophyll, the deformation of velocity fields, and spatial variability. In addition, human expertise is indispensable to provide experiential threshold values for physical parameters to capture the evidence of the eddy. As aforementioned, AI techniques, especially deep learning models, are increasingly employed for remote sensing applications [31]–[33] and are regarded as a potential relevant candidate to reach breakthrough innovations for oceanographic applications [18], [32], [34], [35]. Nevertheless, existing deep learning techniques for eddy detection neglect the importance of capturing accurate edge information of the eddies. The main reason is that most of the deep neural network structures are designed to adapt to the texture feature extraction of natural images. Therefore, it is critical to design a domain-specific deep learning model for eddies detection with satellite remotely sensed sea surface height (SSH) images. In fact, boundary information of eddies could be extracted from the SSH since it is calculated through altimetry satellites that measure the distance between their orbit altitude and the surface of the water. In this work, we propose an eddy detection model via an encoder–decoder way. The rich semantic information can be formed in the encoder module with atrous convolutions. Then, the encoder module sends the low-level features through a skip connection to the decoder. Thus, the decoder module allows for a detailed boundary recovery of ocean eddies, which can then be used to distinguish eddies.

Tracking eddies undoubtedly is another vital task for oceanography. In general, an eddy can move or change its status quickly among several days. Previous eddy tracking algorithms [9], [36]–[39] have achieved success. However, they highly dependent on the given and fixed parameters [19], [40], which means that thresholds should be defined precisely [38]. Generally, eddy tracking follows the results of the detection task to catch its trace. This means that the traces of eddies are formed with the detection results of several successive days. In this work, we further propose a new tracking algorithm that is capable of tracking multiple eddies without relying on tuning physical parameters.

Proliferation of training data acts as the main catalyst for the success of deep learning algorithms. However, tedious and inefficient data annotating presents an obstacle to the development of domain-specific deep learning models. The lack of training data leads to overfitting issues and also makes it hard to get distinctive features [41]–[43]. In contrast to abundant natural images [44], training data for eddy detection are scarce [18], [20]. In order to introduce advanced deep learning approaches into the field of eddy detection, a domain-specific data set is necessary. In this work, we create and provide a public data set called SCSE-Eddy to evaluate the performances of deep learning-based eddy detection methods. The data set contains 15-year-daily SSH images of the South China Sea and its eastern sea (SCSE) areas, which are obtained from the Copernicus Marine Environment Monitoring Service (CMEMS). We believe that the size of this data set is suited to applications involving deep learning model training.

The main contributions of this article are as follows.

1) We propose a domain-specific deep learning model for multieddies detection. It is challenging to distinguish eddies close to each other. One advantage of our model is that it can extract the deep features of eddy boundaries and fit the gap between semantic information and SSH.
2) We design a tracking algorithm to track eddies from data of consecutive days. It captures the key characters of eddies and provides oceanographic visualizations of the dynamical processes governing an eddy movement.
3) We publish a benchmark data set called SCSE-Eddy to evaluate the effectiveness and efficiency of AI-based eddy detection methods. We carry out experiments on the SCSE-Eddy data set and achieve state-of-the-art performance compared with other existing techniques.

The rest of this article is organized as follows. Section II summarizes related works. Section III describes our SCSE-Eddy data set. Section IV presents the detection model and the tracking algorithm. Section V reports the experiments conducted in this work and analyzes the results. Finally, we conclude our study in Section VI.

II. Related Work

Most of the ocean eddy detection approaches depend on the *a priori* physical knowledge of experts. Recently, automatic eddy detection and tracking methods have drawn increasing attention, following a series of works using machine learning techniques for oceanic phenomena discovery and understanding [32], [34], [35]. In this section, we briefly review the most related approaches to our work.

A. Eddy Detection Methods

Traditional algorithms for eddy detection can be divided into four categories: geometric-based [8], [9], [45]–[47], physical parameter-based [4], [10], [11], [48], handcrafted feature-based [12], [13], [49], and hybrid methods [14], [15], [50]. Geometric-based methods are becoming increasingly popular over the past ten years due to their simplicity, and they assume that eddies can be defined as features characterized by a roughly circular shape around their cores. However, they cannot detect eddy details from the data without clear geometrical features. Physical parameter-based methods take full advantage of the information stating that an eddy center is a local extremum. However, they need a set of suitable thresholds for a specific region of interest. Thus, they are task-specific and limited regarding generalization capability.

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Handcrafted feature-based methods design low-level features with experts' prior experience and knowledge and are incapable of representing the dynamics of ocean eddies [20]. Although hybrid methods could be seen as taking the best of both worlds, they still require expert knowledge and noise-free history data to properly execute the detection.

Deep learning-based detection algorithms have already been proposed to discover ocean phenomena, such as ocean eddies [18] and fronts [32]–[35]. Lguensat *et al.* [18] proposed a U-net like encoder–decoder structure for feature extracting. Franz *et al.* [19] utilized the encoder–decoder with a traditional algorithm [51] to detect and track eddies. However, both techniques cannot fully utilize semantic information and features to distinguish eddies' boundaries. Multiscale features fusion was also employed for ocean eddy detection on synthetic aperture radar (SAR) images [20]. Coincidentally, the method in [21] employed PSPNet [52] to fuse multiscale features on different layers, which is excellent at detecting small-scale eddies. Even though rich semantic information is encoded in the last feature map, detailed eddy boundary information is missing due to the pooling and convolutions with the striding operations in the network backbone. In this work, we will focus on integrating the semantic information, multiscale features, and contour details to capture an accurate shape of eddies.

B. Eddy Tracking Algorithms

Ocean eddies tracking, which is a further step to characterize the life cycle of the eddy, is also an important issue in oceanography. Chen *et al.* [39] divided the South China Sea into four regions and provided the genesis and dissipation positions on tracking traces for about two classes of eddies, i.e., cyclonic and anticyclonic eddies. Chelton *et al.* [9] investigated next time-step eddies on predefined elliptical search areas and terminated tracking when the eddy once disappeared. Multiple hypothesis assignment (MHA) [36] extended traditional multiple hypothesis tracking for the cases where the features tracked are noisy or uncertain and uses contextual spatiotemporal information in the presence of noise. With the ocean color imagery, Marcello *et al.* [37] proposed a way for anticyclonic eddy segmentation and tracking, which first incorporates various segmentation and region growing techniques. Faghmous *et al.* [38] tracked eddies when its features were identified by associating a feature in one time step to the next one. Besides, they used only one day as the tolerance to process the problem of ghost eddies. Matsuoka *et al.* [40] proposed a tracking approach that follows an eddy classification system and tracks the detected eddies that vary over time to discover events, such as eddy amalgamation and bifurcation as well as the interaction between eddies and ocean currents.

From a deep learning perspective, long short-term memory (LSTM) architecture, a popular class of recurrent neural networks (RNNs), was introduced to track ocean eddies in [19]. LSTM is capable to fuse past information with current and track ocean eddies. However, the original LSTM structure can hardly be used for the eddy tracking task, due to its high computational cost and also because the result shows that the model cannot distinguish different types of eddies, i.e., cyclone and anticyclone. To address the problem of eddy tracking and release the expert knowledge acquisition, we design a tracking algorithm to track eddies (distinguishing cyclone and anticyclone) on successive days. It displays the coordinates for each eddy with clear edge to gain an excellent tracking result and a precise exhibition of eddies' location.

C. Training Data Set for AI-Based Models

Successful training of deep learning models is constrained by the availability of large and high-quality training data sets. Chelton *et al.* [9] published an ocean eddy trajectories data set that sampled weekly from October 1992 to December 2008. Faghmous *et al.* [38] delivered a data set that includes global daily data from 1993 to 2014 and about 45 million mesoscale features. Tian *et al.* [53] built two global daily mesoscale ocean eddy data sets, which contains 688 thousand eddies that persist for at least four weeks over the 1993–2016 period. The properties of their eddy contours, such as the amplitude, radius, area, and geostrophic current speed, can be transformed into ground-truth labels to train AI-based models while leveraging extra expertise knowledge in physical oceanography. Although these data sets are valuable for eddy-related research, they are not adapted for pixel-level semantic segmentation that is used to training deep learning models.

Several strategies were used in the deep learning eddy classification and tracking literature to construct training data sets. EddyNet [18] is trained on a data set created by a traditional eddy detection method, namely, py-eddy-tracker (PET) [54] on SSH images from 1998 to 2011. Multiscale feature fusion method [20] carries out experiments with SAR images generated by ENVISAT and ERS-2 Satellites between 2005 and 2011. Franz *et al.* [19] used high-resolution sea-level anomaly (SLA) maps around Australia and the South Equatorial current, which are preprocessed products from the ESA Climate Change Initiative (ESACCI). The data set covers daily data from January 1993 until December 2014 with a spatial resolution of 0.25°. For training, they use a traditional parameter-based algorithm, i.e., Okubo-Weiss method [55], [56] which is prone to the noise of data and too subjective to define the thresholds [38]. Duo *et al.* [22] used eddy samples annotated by marine experts to train the learning models. Nevertheless, inefficient labels from experts make it difficult to train an effective model. The PSPNet method [21] used the data from CMEMS. It consists of five-year-daily SSH data, which mainly focuses on the North Pacific Subtropical Countercurrent. Besides, the geometric-based algorithm (VG) [8] is implemented to generate the ground truth [21]. Unfortunately, most data sets that suitable for deep learning methods are not public for researchers. In order to meet the requirement of a large training data set, this work builds a new eddy detection data set that is available to public.

III. Data

This work first presents a large data set (SCSE-Eddy) for multieddies detection, which covers the SCSE areas, as shown

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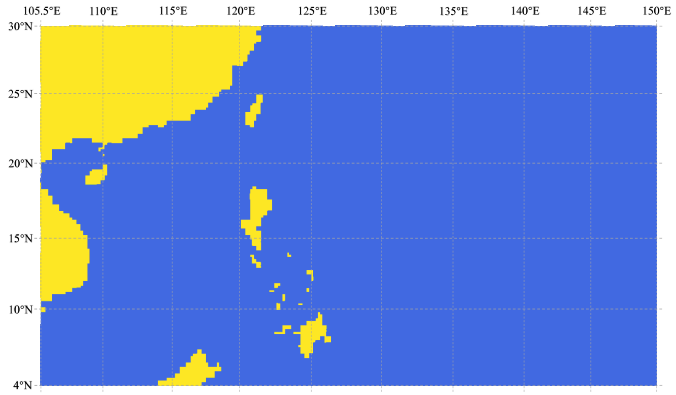


Fig. 2. SCSE-Eddy covers the whole SCSE areas. The longitude of this area is from 105.5°E to 150°E, and the latitude is from 4°N to 30°N. Yellow region represents land or island.

in Fig. 2. SCSE-Eddy contains 15 years of data from September 23, 2003 to September 23, 2018, resulting in 5480 daily images for multieddies detection.

In order to create the ground-truth labels, a popular method, namely PET [54], is employed to generate eddies. Then, we create pixel-level annotation for deep learning methods. PET is closely related to CSS11 [9] with many remarkable differences such as eddy scales and ensuring one local extremum of an eddy. Since the quality of our supervised learning technique will heavily depend on the quality of PET outputs, we called for the assistance of professional oceanographers to validate the ground-truth labels. Ambiguous annotation for eddies will be corrected or removed manually.

The SCSE-Eddy data set is divided into training and validation sets. The training set is formed with the first 15-year data (i.e., 4750 images), whereas the validation set is the last two-year (i.e., 730 images). Each piece of data is represented as a one-channel image with a spatial resolution of 0.25° and the size is 302 × 184. The data set is publicly available on the following GitHub repository https://github.com/ouc-ocean-group/EddyData.

IV. Methodology

As shown in Fig. 1, our framework focuses on two aspects: multieddies detection and automatic eddies tracking. The rich semantic information is formed in the encoder module with atrous convolutions. The decoder module allows detailed boundary recovery for ocean eddies. Our tracking algorithm is specially designed to take full use of the results from deep learning-based eddy detection methods.

A. Deep Detection Model

We propose a deep learning-based eddy detection model to capture the semantic and accurate boundary context as well as multiscale features. An overview of our model is shown in Fig. 3, and it mainly consists of two modules, i.e., an encoder and a decoder. The encoder module is based on a ConvNet-based feature extractor that yields primary features of input SSH data. It includes a modified Xception backbone [57] and several atrous convolutional operations [58]. Convolution operation aims to extract and integrate the deep features on the image with convolution kernels. Convolution kernels are equivalent to a feature filtering matrix, which learns relevant features. Atrous convolution can enlarge the field of view of filters to incorporate larger context and is formulated as follows:

$$y(i) = \sum_{k} x(i + r * k)w(k) \tag{1}$$

where i denotes the location on the output feature map y, w is the convolutional filter, x represents the input feature map, and r determines the stride. The feature extractor encodes multiscale contextual information via atrous convolution at multiple scales [59]. Subsequently, the decoder module is designed to recover the edges of ocean eddies and aims to keep an accurate boundary context. Primary features with multiscale contextual information ensure that our model is capable of extracting fundamental characteristics of eddies. The eddies' boundaries can be recovered after the features were fully extracted.

Xception is employed as the backbone for our proposed architecture. The original Xception architecture [57] is a linear stack of depthwise separable convolution layers with residual connections. Specifically, we reduce the layers in the middle flow while doubling the repeat times compared with the original Xception. The main reason is that eddies are smaller when exhibited on the images. Too many layers in one flow lead to the fact that the model cannot detect eddies with small radii. Besides, max-pooling operations are replaced by depthwise separable convolutions with striding except for the last layer. Finally, batch normalization [60] and ReLU activation are added after each 3 × 3 depthwise convolution. A sketch of our modified Xception is shown in Fig. 4.

Moreover, the encoder module also applies several parallel atrous convolutions with different rates (called atrous spatial pyramid pooling (ASPP) [61], [62]), which probes convolutional features at multiple scales. We use the last feature map as the encoder's output. Besides, we extract features at an atrous rate of 6, 12, and 18, while for the decoder module, the encoder features are bilinearly upsampled by a factor of 4 and then concatenated with the corresponding low-level features from the network backbone with the same spatial resolution. The following formula is used to calculate the upsampled feature value at current position (x, y):

$$\begin{aligned} f(x,y) \approx r * \Bigg(& \frac{f(A_{11})}{(x_2-x_1)(y_2-y_1)}(x_2-x)(y_2-y) \\ & + \frac{f(A_{21})}{(x_2-x_1)(y_2-y_1)}(x-x_1)(y_2-y) \\ & + \frac{f(A_{12})}{(x_2-x_1)(y_2-y_1)}(x_2-x)(y-y_1) \\ & + \frac{f(A_{22})}{(x_2-x_1)(y_2-y_1)}(x-x_1)(y-y_1) \Bigg) \end{aligned} \tag{2}$$

where $f(A_{11})$, $f(A_{12})$, $f(A_{21})$, and $f(A_{22})$ are the given values, $A_{ij} = (x_i, y_j)$, $i, j \in \{1, 2\}$, and r represents the

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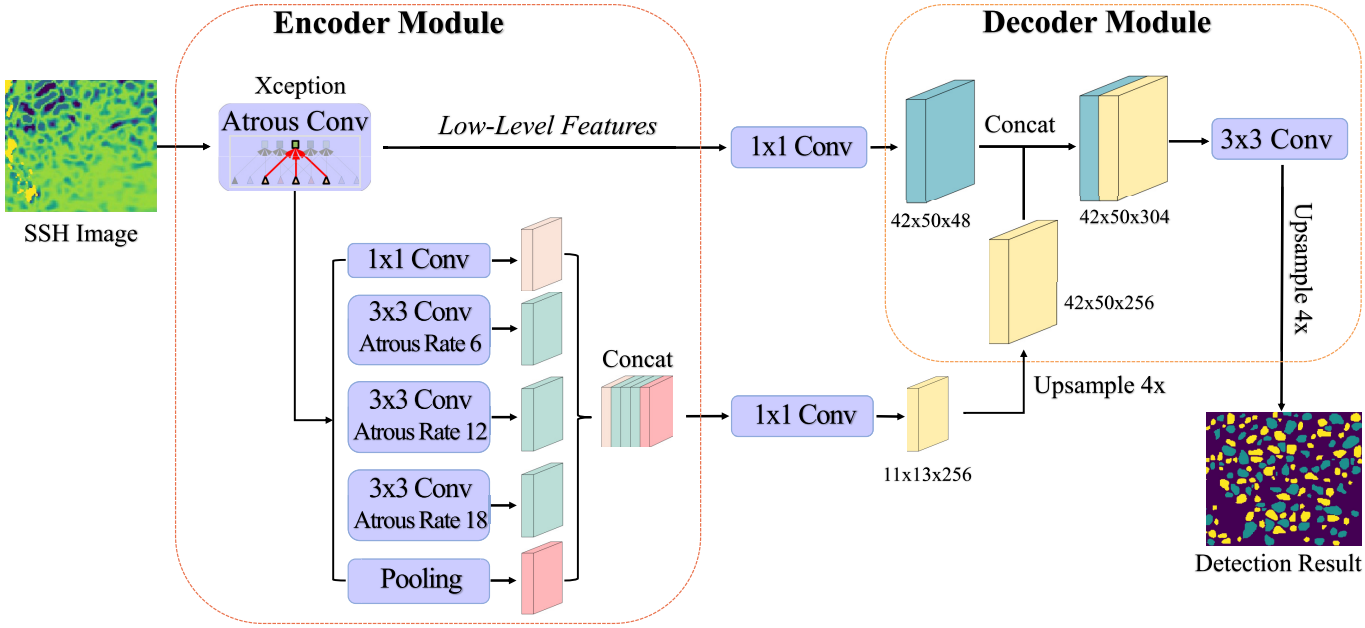


Fig. 3. Overview framework of our multieddies detection model. It takes the satellite sea SSH data as input. The output is the detected eddies, including cyclones and anticyclones with different colors.

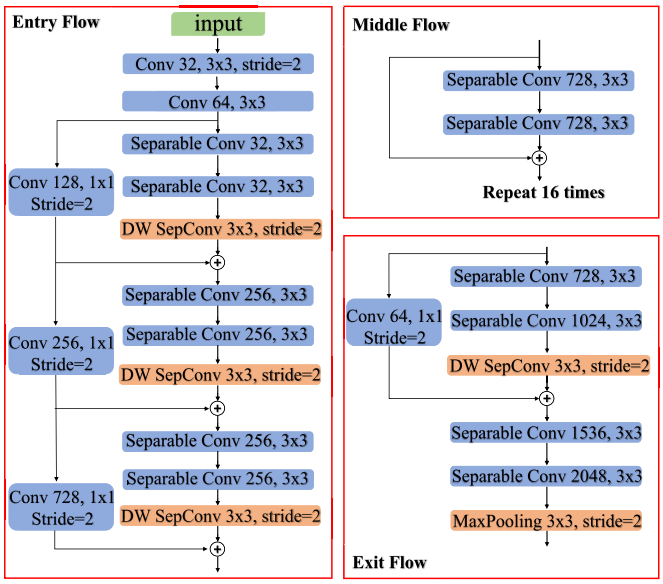


Fig. 4. Sketch of the modified Xception. The entry flow replaces the max-pooling operations with depthwise separable convolutions with striding and extra batch normalization. The middle flow has only a few layers. The last separable convolution in exit flow is followed by a max-pooling layer.

upsample rate. Another 1×1 convolution on the low-level features is utilized to reduce the number of channels. Because the corresponding low-level features usually contain a large number of channels which out-weight the importance of rich encoder features [63]. Finally, we apply a few 3×3 convolutions to refine the features followed by a last bilinear upsampling with a factor of 4.

B. Tracking Algorithm

Ocean eddies may persist for periods of days to months, even years. Eddy tracking is an important issue for eddy-related research. To gain an understanding of ocean eddy evolution, tracking eddy is necessary in order to predict its movement. In the last decade, several eddy tracking methods [19], [37], [40] have been proposed. In this work, we further propose a robust Median blur, Contours finding, Moments computation, Location (MCML) tracking algorithm to track ocean eddies and distinguish different types of eddies. Our MCML tracking algorithm is easy to implement and has a low computational cost.

The MCML tracking algorithm imports the eddy detection results to track multieddies simultaneously. At first, given the original image $f(x, y)$, a median blur kernel is applied to the images to preserve the edge information as follows:

$$g(x, y) = \text{Median}[f(x - j, y - k)] \tag{3}$$

where $j, k \in w$, w stands for the input template and $g(x, y)$ is the result of blurred image. Second, we find all the points that make up the boundary with a value of 1 and then detect the closed contours formed by these points. Eddy contours are represented as $K = \{k_1, k_2, \ldots, k_m\}$, where m denotes the number of detected eddies. Third, we calculate the moments μ_{pq} for each contour using the following formulas:

$$\mu_{pq} = \sum_i \sum_j (i - \bar{i})^p (j - \bar{j})^q f(i, j) \tag{4}$$

$$m_{pq} = \sum_i \sum_j i^p j^q f(i, j). \tag{5}$$

where $f(i, j)$ is the gray value at the position (i, j), p and q are the orders, and $\bar{i} = m_{10}/m_{00}$ and $\bar{j} = m_{01}/m_{00}$. As a matter of fact, m_{00} stands for gray mass of image, and the cores (X_j, Y_j) of K are equivalent to $(\bar{i_n}, \bar{j_n})$. We further label the eddies with "cyclone" and "anticyclone" according to the detection results. Moreover, if the eddy's core of next time step is found on a 5×5 pixel space around current

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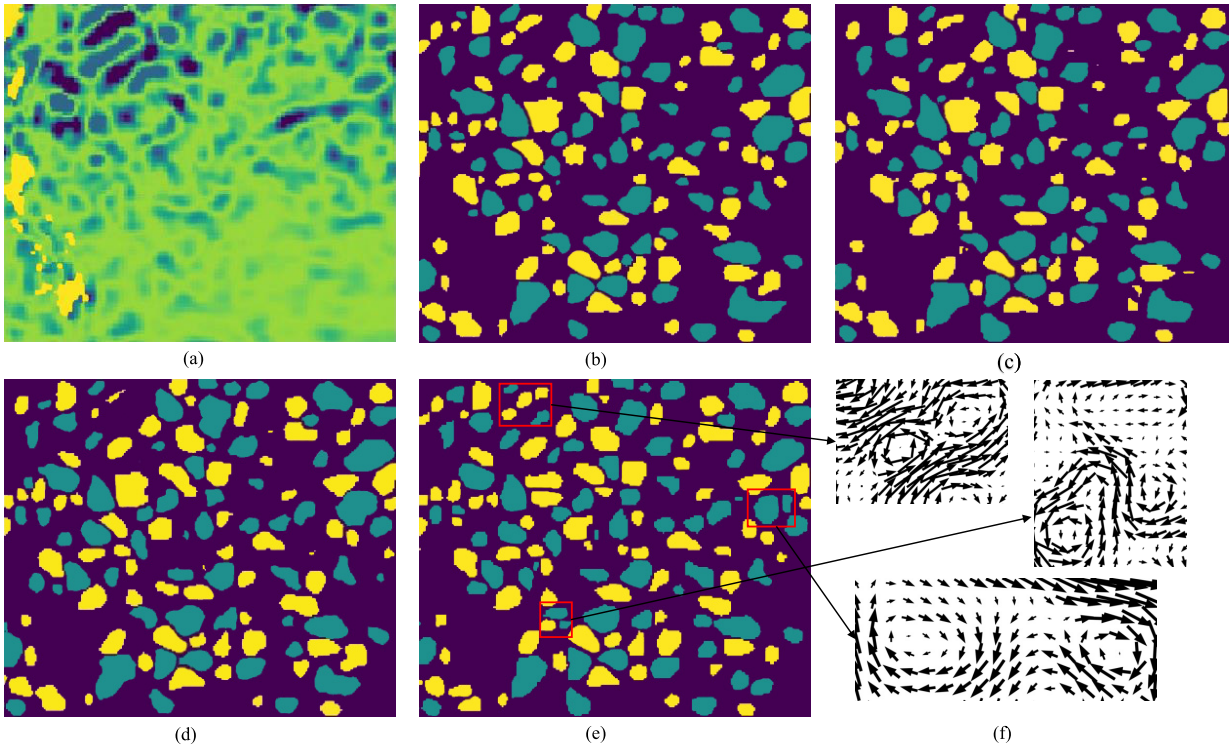


Fig. 5. Illustration of the comparison results with the state of the arts. (Bottom-right corner) Geostrophic flows of three small sensitive regions. (a) SSH. (b) PET method. (c) EddyNet. (d) PSPNet-based. (e) Ours. (f) Geostrophic flows.

Algorithm 1 MCML: Ocean Eddy's Tracking and Location

Input:
Image $I_1, I_2, \ldots, I_n$ with detected eddies where n denotes the number of days;
Output:
S: Traces of ocean eddies;
$S = \varnothing$;
for $i = 1$; $i = i + 1$; $i < n$ **do**
  transfer I_i into I_i' by median blur as Eq.(3);
  detect closed contours $K = \{k_1, k_2, \ldots, k_m\}$ on I_i';
  for $j = 1$; $j = j + 1$; $j < m$ **do**
    compute the moments of I_i' by Eq.(4);
    search for the core (X_j, Y_j) of k_j by Eq.(5);
    set core-label = 'cyclone' or 'anticyclone';
    display coordinate of (X_j, Y_j) on I_1';
    append (X_j, Y_j) to S;
  end for
end for
I_1' illustrates the traces of eddies.

one, we treated them as the same one. Finally, we display the coordinates of the core of each eddy. The trace of the eddy clearly shows its location and variation. Implementation details are demonstrated in Algorithm 1.

V. Experiments

In this section, we describe the experiments conducted in this work. We will show the eddy detection results compared with several state-of-the-art methods and tracking results. Our code is open source and can be downloaded from the following GitHub repository: https://github.com/ouc-ocean-group/EddyData.

A. Experiment Design and Details

In our comparison experiments, all deep learning-based methods are trained on the SCSE-Eddy data set without any augmentation. The codes were written using Keras deep learning library [64]. All the methods are trained with 300 epochs and a batch size of 8. Experiments are conducted on one RTX-2080Ti GPU. MCML tracking algorithm uses the results of the detection task and is capable to track eddies for several days. Aiming for a better illustration, we will track eddies for five days and display their traces on the map.

B. Detection Results

Fig. 5 shows the detection results of PET [54], EddyNet [18], PSPNet [21], and our proposed method. We can see that our method produces a similar false-color composite image with PET because our model was trained with its results. However, compared with others, our method can discover small and obscure eddies accurately, as shown in Fig. 5(f). The data from CMEMS contain "ugos" and "vgos" attributes that stand for eastwards and northwards flows, respectively. We can visualize them as geostrophic flows to validate the results. PSPNet-based method is also capable of identifying more eddies, especially for small ones (i.e., with small radii). The advantage of our method is that we are able to capture

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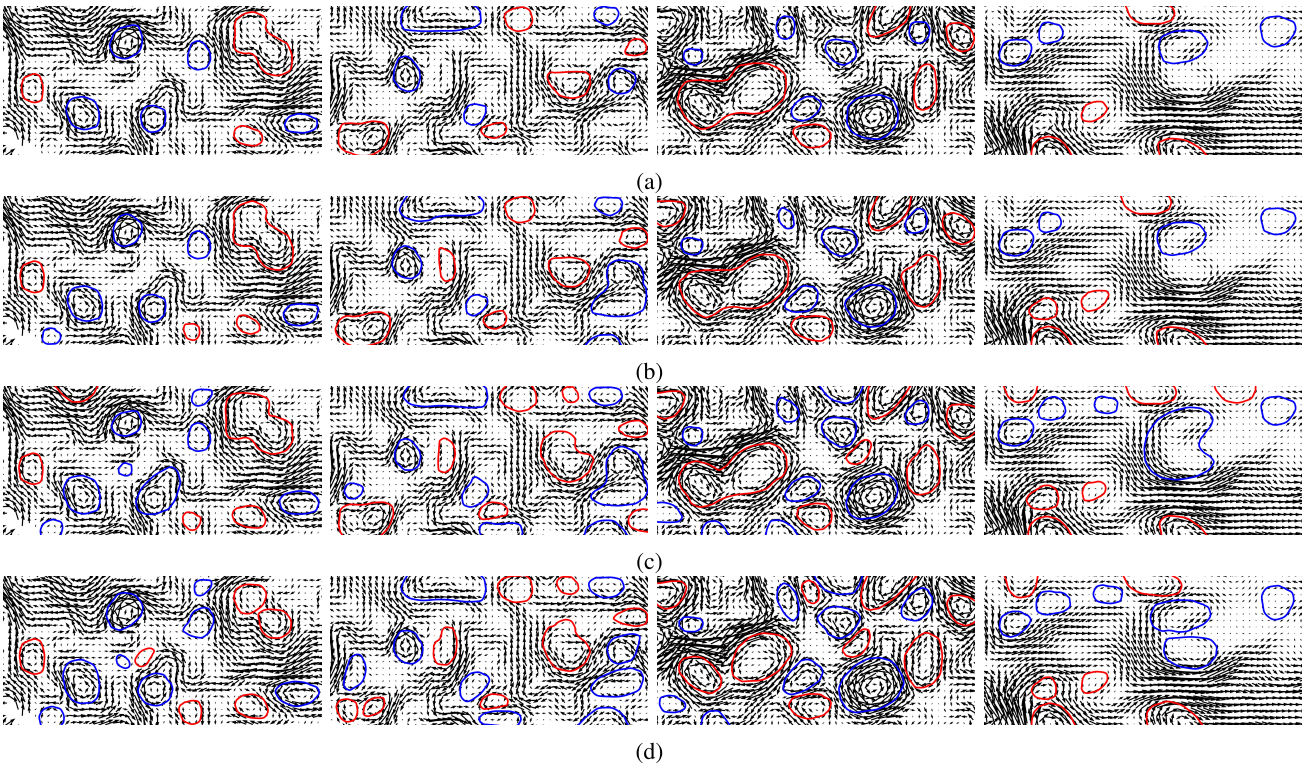


Fig. 6. Visualization of the detection results on a small region with geostrophic flows. Red and blue contours stand for anticyclones and cyclones, respectively. (a) PET. (b) EddyNet. (c) PSPNet. (d) Ours.

accurate edges of eddies. Hence, eddies that are close to each other can be separated and recognized by our method, as shown in Figs. 5(f) and 6.

Eddies appear generally more "roundish," nearly elliptic, rather than irregular polygon due to the Coriolis force [65], [66]. We find that our results are more realistic than others. We visualize some situations for analysis of the detection results on small regions from the geostrophic flows, as shown in Fig. 6. Columns from the left to right stand for detection results sampled on December 26, 2005, June 21, 2010, March 5, 2014 and September 6, 2018. We can see that our method can capture the edge information well, especially for eddies that are close to each other. Therefore, our method is superior to others in detecting small-scale eddies. Also, we can find that PSPNet method is good at detecting the boundary eddies, as shown in the fourth column of Fig. 6. Here, the boundary eddy means that the half eddy appears at the edge of the study area, due to the data selection [21].

Radius, as a critical property of ocean eddy, is a valuable measure for the detection performance. We analyze the radii distribution of eddies as shown in Fig. 7, which indicates that the radius range of 30 to 50 km covers the utmost proportion. The number of eddies with the radii under 70 km detected by our method is larger than PSPNet, which means that it performs better than PSPNet on small-scale eddy detection.

AVISO+ (www.aviso.altimetry.fr) provides a mesoscale eddy atlas product, which includes eddy's radius. We show the comparison results of various detection methods in Table I. We can see that deep learning methods as well as PET algorithm are capable of detecting these eddies cover vast sea area.

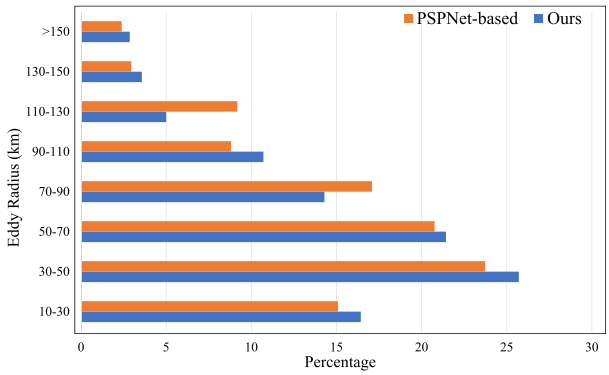


Fig. 7. Radius distribution of the eddies detected by PSPNet and ours.

TABLE I
COMPARISON OF RADIUS DETECTED BY FIVE METHODS

Methods	AVISO+	PET	EddyNet	PSPNet	Ours
Mean Radius (km)	99.58	84.75	87.29	90.63	93.81
Max Radius (km)	267.00	298.38	298.81	297.57	300.48

We record the number of cyclones and anticyclones detected by different methods in Table II. N-Cyclone, N-Anticyclone, and N-All stand for the number of cyclones, anticyclones, and all eddies, respectively. Generally, deep learning methods, such as EddyNet, PSPNet, and ours, discover more eddies than the traditional PET method. As discussed in [21], deep learning models have the ability to detect small-size eddies.

Here, we further conduct experiments to investigate the performance of different deep learning-based methods. We test

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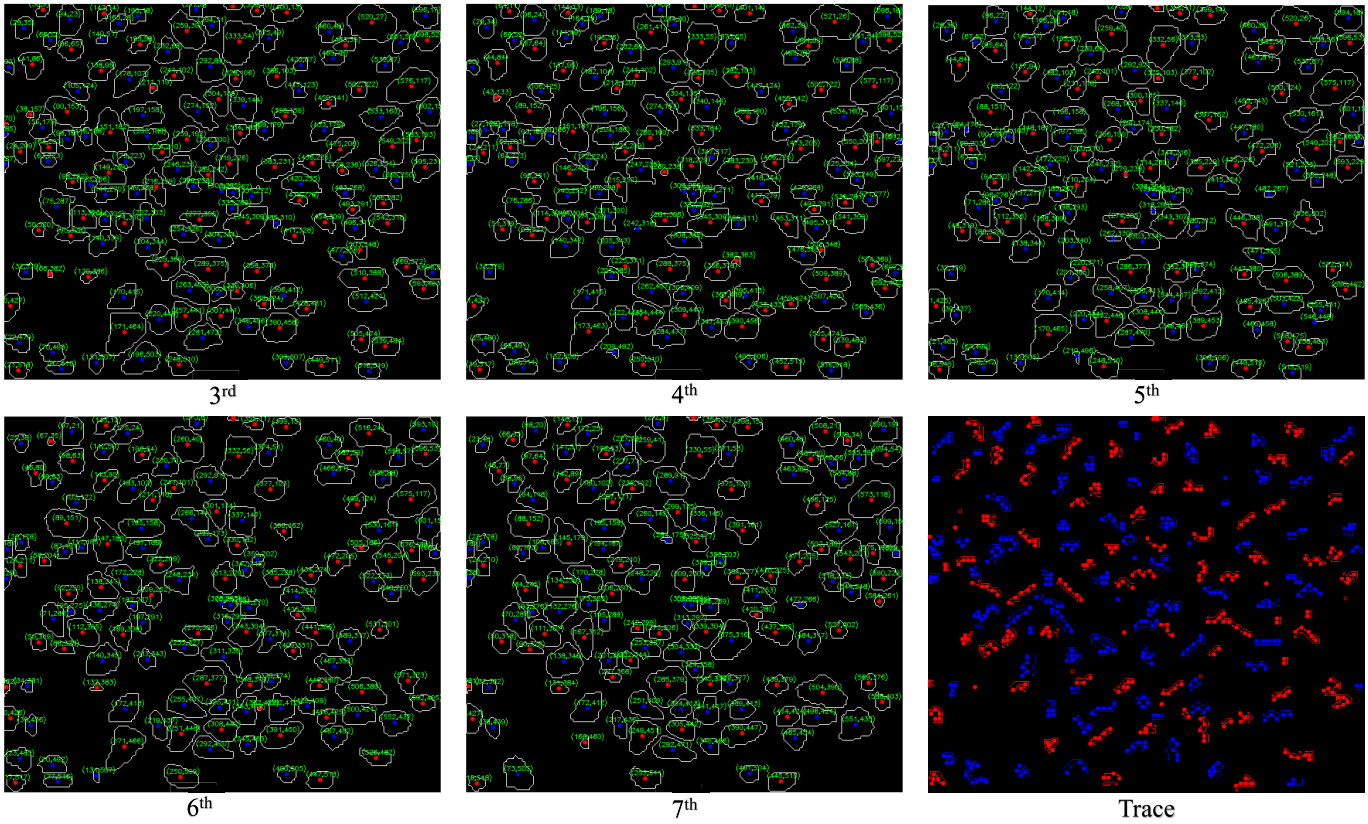


Fig. 8. Results of MCML algorithm from October 3–7, 2016. The coordinates of each core can be displayed for visualizing the location of each eddy.

TABLE II
NUMBER OF EDDIES DETECTED WITH VARIOUS METHODS

Methods	N-Cyclone	N-Anticyclone	N-All
PET [54]	4473 (54.04%)	3737 (45.96%)	8210
EddyNet [18]	4502 (54.27%)	3794 (45.73%)	8296
PSPNet [21]	4614 (54.15%)	3906 (45.85%)	8520
Ours	**4662** (54.12%)	**3952** (45.88%)	**8614**

TABLE III
EDDY DETECTION PERFORMANCES OF DEEP LEARNING-BASED METHODS

Methods	Recall	Execution Time (Sec)
EddyNet	92.34%	108.13
PSPNet	93.83%	82.57
Ours	94.81%	55.06

all the methods on the SCSE-Eddy validation set. As shown in Table III, our method can detect most of the eddies compared with others. Moreover, execution time represents the processing time on 730 images in the validation set. Table III also shows the superiority of our method in execution time.

C. Tracking Results

To illustrate the performance of our tracking method, we capture the traces of eddies for five days. As shown in the previous section, our detection model achieves promising performance. Therefore, we will track the eddies that are discovered by our detection model. The detection model will be implemented as the input module of our MCML tracking algorithm. Specifically, the MCML algorithm devotes to finding the contours of all eddies and labeling their cores. As shown in Fig. 8, we assign different colors to distinguish the cyclone (blue) and anticyclone (red). The coordinates of each core are displayed for clear visualizing the location of each eddy. We chose the SSH image data randomly for the purpose of testing our algorithm's ability and proving its robustness. Fig. 8 shows the results of eddy tracking from October 3, 2016 to October 7, 2016. The trace of each eddy indicates the direction and trail of its motion. Some cores are hollow, which means that these eddies do not exist on the early tracking days and we represent their location at the first occurrence.

From the first generation of the eddy to its disappearance, the whole process is called the lifetime of an ocean eddy. The histogram of eddy lifetimes is shown in Fig. 9, which depicts the number of eddies with regard to the lifetime. We can see that the distribution of cyclonic and anticyclonic eddies is similar, which is also confirmed by some previous research works [9], [67].

All eddies are moving constantly during their lifetime. The longer of a lifetime, the longer distance eddies move. Fig. 10 shows the cyclonic and anticyclonic ocean eddies' average moving distance with the lifetime from 1 month to 12 months. Table II and Fig. 9 show that the number of cyclones is higher than that of anticyclones; however, Fig. 10 shows that the average moving distance of anticyclones is much longer than

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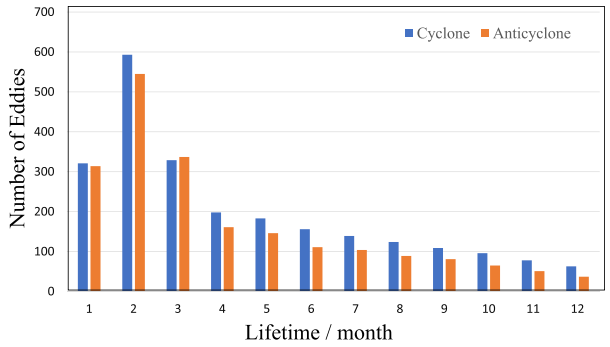


Fig. 9. Histograms of the lifetimes (1–12 months) of the cyclonic and anticyclonic eddies.

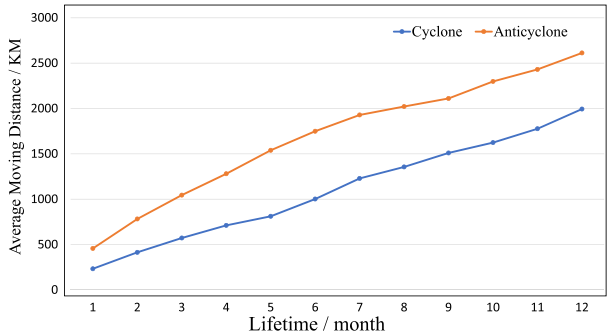


Fig. 10. Average moving distances of cyclones and anticyclones with regard to the lifetime.

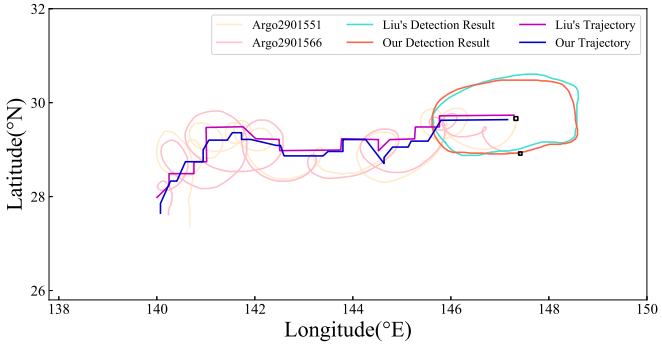


Fig. 11. Comparison of eddy trajectory of our tracking algorithm with *in situ* data.

that of cyclones. This means that anticyclonic eddies seem to be more active than their cyclonic counterparts.

To better validate the effectiveness of our method and show the tracking result clearly, we carry out experiments on *in situ*data and compare the tracking results with Liu *et al.* [45], as shown in Fig. 11. The black rectangles represent argos and are connected with their trajectories. The argos' lifetime is from March 28, 2014 to September 30, 2014, while the eddy's we detected is two days longer and ended on October 2, 2014. We show the comparison of detecting as well as tracking results of an anticyclonic eddy, which can be seen that our tracking algorithm matches argos' trajectories better than Liu's method.

VI. Conclusion

Automatic detection and tracking of ocean eddies are important consequences of the advances in satellite altimetry over the past two decades. In this article, we focus on the challenge of ocean eddies' automatic detection and tracking problems via the use of AI techniques. To capture accurate boundary information for eddy detection, we propose a convolutional neural networks model to fit the gap between semantic information and SSH. We achieve promising results and show superiority compared with state-of-the-art eddy detection methods. Then, our proposed MCML tracking algorithm is used to capture the movement and location of eddies following the detection module. Our method is shown to yield excellent tracking performance and reveals several insights about the differences between cyclonic and anticyclonic eddies. Furthermore, we make the SCSE-Eddy data set available to help other researchers evaluate the performances of their AI-based eddy detection and tracking methods. We believe that our methods are a valuable complement to the state-of-the-art techniques and can provide new answers to questions investigated by the ocean remote sensing and physical oceanographic communities. A key problem with deep learning is that they are generally seen as being "black boxes": while it may work well in particular applications without understanding how it works, leading to challenges in improving the performance when fails. The black boxes learning may cause the trust issues of oceanographers. In the future work, we will pay attention to investigating new deep learning methods for eddy detection and tracking with contextual explanation.

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